

# Øvingsoppgave 7

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# CHAPTER

## 1.1 Oppgave 1

Regn ut følgende ubestemte integraler ved hjelp av substitusjon:

a)  $\int 9(4x - 3)^8 dx$

$$u = 4x - 3$$

$$\frac{du}{dx} = 4 \quad | \cdot \frac{dx}{4}$$

$$\frac{1}{4} du = dx$$

$$\int 9(4x - 3)^8 dx = \int 9(u)^8 \cdot \frac{1}{4} du = \frac{9}{4} \int (u)^8 du$$

$$\frac{9}{4} \int (u)^8 du \quad | \int x^r dx = \frac{1}{r+1} x^{r+1} + C$$

$$\frac{9}{4} \cdot \frac{1}{9} u^9 + C \quad | \frac{9}{4} \cdot \frac{1}{9} = \frac{1}{4}$$

$$\boxed{\boxed{\int 9(4x - 3)^8 dx = \frac{1}{4}(4x - 3)^9 + C}}$$

b)  $\int e^{2x} dx$

$$u = 2x$$

$$\frac{du}{dx} = 2 \quad | \cdot \frac{dx}{2}$$

$$\frac{1}{2} du = dx$$

$$\int e^{2x} dx = \int e^u \cdot \frac{1}{2} du = \frac{1}{2} \int e^u du$$

$$\frac{1}{2} \int e^u du \quad | \int e^x dx = e^x + C$$

$$\boxed{\boxed{\int e^{2x} dx = \frac{1}{2}e^{2x} + C}}$$

c)  $\int \cos(4x) dx$

$$u = 4x$$

$$\frac{du}{dx} = 4 \quad | \cdot \frac{dx}{4}$$

$$\frac{1}{4} du = dx$$

$$\int \cos(u) dx = \int \cos(u) \frac{1}{4} du = \frac{1}{4} \int \cos(u) du$$

$$\frac{1}{4} \int \cos(u) du \quad | \int \cos(x) dx = \sin(x) + C$$

$$\boxed{\int \cos(4x) dx = \frac{1}{4} \sin(4x) + C}$$

d)  $\int x e^{x^2-1} dx$

$$u = x^2 - 1$$

$$\frac{du}{dx} = 2x \quad | \cdot \frac{dx}{2x}$$

$$\frac{1}{2x} du = dx$$

$$\int x e^u dx = \int x e^u \cdot \frac{1}{2x} du$$

$$\int x e^u \cdot \frac{1}{2x} du \quad | x \cdot \frac{1}{2x} = \frac{x}{1} \cdot \frac{1}{2x} = \frac{1}{2}$$

$$\frac{1}{2} \int e^u du \quad | \int e^x dx = e^x + C$$

$$\boxed{\int x e^{x^2-1} dx = \frac{1}{2} e^x + C}$$

e)  $\int (x-1) \sin(x^2 - 2x + 1) dx$

$$u = x^2 - 2x + 1$$

$$\frac{du}{dx} = 2x - 2 \quad | \cdot \frac{dx}{2x-2}$$

$$\frac{1}{2x-2} du = dx \quad | 2x-2 = 2(x-1)$$

$$\int (x-1) \sin(u) \cdot \frac{1}{2(x-1)} du \quad | (x-1) \cdot \frac{1}{2(x-1)} = \frac{1}{2}$$

$$\frac{1}{2} \int \sin(u) du \quad | \int \sin(x) dx = -\cos(x) + C$$

$$\frac{1}{2} (-\cos(x)) + C = -\frac{1}{2} \cos(x) + C$$

$$\boxed{\boxed{\int (x-1) \sin(x^2 - 2x + 1) dx = -\frac{1}{2} \cos(x) + C}}$$

f)  $\int \frac{\sin(\sqrt{x})}{\sqrt{x}} dx$

$$u = \sqrt{x} = x^{\frac{1}{2}}$$

$$\frac{du}{dx} = \frac{1}{2} x^{-\frac{1}{2}} \quad | \cdot \frac{dx}{\frac{1}{2} x^{-\frac{1}{2}}}$$

$$\frac{1}{\frac{1}{2} x^{-\frac{1}{2}}} du = dx \quad | \frac{1}{\frac{1}{2} x^{-\frac{1}{2}}} = \frac{x^{\frac{1}{2}}}{\frac{1}{2}} = 2\sqrt{x}$$

$$2\sqrt{x} = dx$$

$$\int \frac{\sin(u)}{\sqrt{x}} \cdot 2\sqrt{x} du \quad | \frac{1}{\sqrt{x}} \cdot 2\sqrt{x} = 2$$

$$2 \int \sin(u) du \quad | \int \sin(x) dx = -\cos(x) + C$$

$$2 \cdot (-\cos(u)) + C = -2 \cos(u) + C$$

$$\boxed{\boxed{\int \frac{\sin(\sqrt{x})}{\sqrt{x}} dx = -2 \cos(\sqrt{x}) + C}}$$

g)  $\int e^{-3 \cos(x)} \cdot \sin(x) dx$

$$u = -3 \cos(x) \quad | (a \cdot f)' = a \cdot f'$$

$$\frac{du}{dx} = -3(-\sin(x)) = 3 \sin(x) \quad | \cdot \frac{dx}{3 \sin(x)}$$

$$\frac{1}{3 \sin(x)} du = dx$$

$$\int e^u \cdot \sin(x) \cdot \frac{1}{3 \sin(x)} du \quad | \sin(x) \cdot \frac{1}{3 \sin(x)} = \frac{1}{3}$$

$$\frac{1}{3} \int e^u du \quad | \int e^x = e^x + C$$

$$\boxed{\boxed{\int e^{-3 \cos(x)} \cdot \sin(x) dx = \frac{1}{3} e^{-3 \cos(x)} + C}}$$

## 1.2 Oppgave 2

Bestem følgende integral:  $\int \frac{5}{\sqrt{4-2x^2}} dx$

$$\int \frac{5}{\sqrt{4-2x^2}} dx \quad | \quad 4-2x^2 = 2(2-x^2) \rightarrow \sqrt{2(2-x^2)} = \sqrt{2} \cdot \sqrt{(2-x^2)}$$

$$\int \frac{5}{\sqrt{2} \cdot \sqrt{(2-x^2)}} dx \quad | \quad \frac{5}{\sqrt{2} \cdot \sqrt{(2-x^2)}} = \frac{5}{\sqrt{2}} \cdot \frac{1}{\sqrt{(2-x^2)}}$$

$$\int \frac{5}{\sqrt{2}} \cdot \frac{1}{\sqrt{(2-x^2)}} dx \quad | \quad \text{Tar konstant ut av integral}$$

$$\frac{5}{\sqrt{2}} \int \frac{1}{\sqrt{(2-x^2)}} dx \quad | \quad \int \frac{1}{\sqrt{a^2-x^2}} dx = \arcsin\left(\frac{x}{a}\right) + C, \quad a^2 = 2 \Rightarrow a = \sqrt{2}$$

$$\boxed{\boxed{\int \frac{5}{\sqrt{4-2x^2}} dx = \frac{5}{\sqrt{2}} \arcsin\left(\frac{x}{\sqrt{2}}\right) + C}}$$

## 1.3 Oppgave 3

Regn ut følgende integral:  $\int \frac{1}{x^2+4x+13} dx$

Finner fullstendig kvadrat av  $x^2 + 4x + 13$ :

$$x^2 + 4x = (x+2)^2 \quad | \quad (x+2)^2 = x^2 + 4x + 4, \quad 13 - 4 = 9$$

$$x^2 + 4x + 13 = (x+2)^2 + 9$$

$$\int \frac{1}{x^2 + 4x + 13} dx = \int \frac{1}{(x+2)^2 + 9} dx$$

$$u = x + 2$$

$$\frac{du}{dx} = 1 \Rightarrow du = dx$$

$$\int \frac{1}{u^2 + 9} du \quad | \quad \int \frac{1}{x^2 + a^2} = \frac{1}{a} \arctan\left(\frac{x}{a}\right) + C, \quad a^2 = 9 \Rightarrow a = \sqrt{9} = 3$$

$$\int \frac{1}{u^2 + 9} du = \frac{1}{3} \arctan\left(\frac{u}{3}\right) + C$$

$$\boxed{\boxed{\int \frac{1}{x^2+4x+13} dx = \frac{1}{3} \arctan\left(\frac{x+2}{3}\right) + C}}$$

## 1.4 Oppgave 4

Regn ut følgende ubestemte integraler ved å benytte delvis integrasjon:

$$\int u'v \, dx = uv - \int uv' \, dx$$

a)  $\int (3x - 2)e^x \, dx$

Velger  $e^x$  som  $u$ .

$$u' = e^x \quad | \quad u = e^x$$

$$v = 3x - 2 \quad | \quad v' = 3$$

$$\begin{aligned} &= e^x(3x - 2) - \int e^x \cdot 3 \, dx \quad | \quad \int 3e^x \, dx = 3 \int e^x \, dx = 3e^x + C \\ &= e^x(3x - 2) - 3e^x + C \\ &= 3e^x - 2e^x - 3e^x + C \\ &= 3xe^x - 5e^x + C \\ &= (3x - 5)e^x + C \end{aligned}$$

$$\boxed{\int (3x - 2)e^x \, dx = (3x - 5)e^x + C}$$

b)  $\int x^2 e^x \, dx$

Velger  $e^x$  som  $u$ .

$$u' = e^x \quad | \quad u = e^x$$

$$v = x^2 \quad | \quad v' = 2x$$

$$= e^x x^2 - \int e^x \cdot 2x \, dx \quad | \quad \text{velger igjen } e^x \text{ som } u$$

$$u' = e^x \quad | \quad u = e^x$$

$$v = 2x \quad | \quad v' = 2$$

$$\begin{aligned} &= e^x x^2 - (e^x \cdot 2x - \int e^x 2 \, dx) \quad | \quad \int e^x 2 \, dx = 2e^x + C \\ &= e^x x^2 - (e^x 2x - 2e^x) + C \\ &= e^x x^2 - e^x 2x + 2e^x + C \\ &= e^x (x^2 - 2x + 2) + C \end{aligned}$$

$$\boxed{\boxed{\int x^2 e^x dx = e^x(x^2 - 2x + 2) + C}}$$

c)  $\int (2x^2 - 3x) \cos(x) dx$

Velger  $\cos(x)$  som  $u$ .

$$u' = \cos(x) \quad | \quad u = \sin(x)$$

$$v = 2x^2 - 3x \quad | \quad v' = 4x - 3$$

$$= \sin(x)(2x^2 - 3x) - \int \sin(x)(4x - 3) dx \quad | \quad \text{Velger } \sin(x) \text{ som } u.$$

$$u' = \sin(x) \quad | \quad u = -\cos(x)$$

$$v = 4x - 3 \quad | \quad v' = 4$$

$$\begin{aligned} &= \sin(x)(2x^2 - 3x) - (-\cos(x)(4x - 3) - \int -\cos(x) \cdot 4 dx) \quad | \quad = 4 \int -\cos(x) dx \\ &= -4 \sin(x) + C \end{aligned}$$

$$\sin(x)(2x^2 - 3x) - (-\cos(x)(4x - 3) - (-4 \sin(x))) + C$$

$$\sin(x)(2x^2 - 3x) - (-\cos(x)(4x - 3) + 4 \sin(x)) + C$$

$$\sin(x)(2x^2 - 3x) + \cos(x)(4x - 3) - 4 \sin(x) + C$$

$$\sin(x)(2x^2 - 3x - 4) + \cos(x)(4x - 3) + C$$

$$\boxed{\boxed{\int (2x^2 - 3x) \cos(x) dx = \sin(x)(2x^2 - 3x - 4) + \cos(x)(4x - 3) + C}}$$

d)  $\int \ln(x + 1) dx$

$$\int \ln(x + 1) dx = \int 1 \cdot \ln(x + 1) dx$$

$$u' = 1 \quad | \quad u = x$$

$$v = \ln(x + 1) \quad | \quad v' = \frac{1}{x + 1}$$

$$x \ln(x + 1) - \int \frac{x}{x + 1} dx \quad | \quad x : x + 1 = 1 - \frac{1}{x + 1}$$

$$x \ln(x + 1) - \int (1 - \frac{1}{x + 1}) dx \quad | \quad \int (1 - \frac{1}{x + 1}) dx = \int 1 dx - \int \frac{1}{x + 1} dx = x - \ln(x + 1) + C$$

$$\begin{aligned}
&= x \ln(x+1) - (x - \ln(x+1)) + C \\
&= x \ln(x+1) - x + \ln(x+1) + C \\
&\quad (x+1) \ln(x+1) - x + C
\end{aligned}$$

$$\boxed{\boxed{\int \ln(x+1) dx = (x+1) \ln(x+1) - x + C}}$$

e)  $\int \arctan(x) dx$

$$\begin{aligned}
\int \arctan(x) dx &= \int 1 \cdot \arctan(x) dx \\
u' &= 1 \quad | \quad u = x
\end{aligned}$$

$$\begin{aligned}
v &= \arctan(x) \quad | \quad v' = \frac{1}{1+x^2} \\
&= x \arctan(x) - \int \frac{x}{1+x^2} dx
\end{aligned}$$

Substitusjon:

$$\begin{aligned}
u &= 1+x^2 \\
\frac{du}{dx} &= 2x \quad | \quad \frac{dx}{2x} \\
\frac{1}{2x} du &= dx \\
&= x \arctan(x) - \int \frac{x}{u} \cdot \frac{1}{2x} du \\
&= x \arctan(x) - \frac{1}{2} \int \frac{1}{u} du \quad | \quad \int \frac{1}{x} dx = \ln|x| + C \\
&= x \arctan(x) - \frac{1}{2} \cdot \ln|u| + C \\
&= x \arctan(x) - \frac{1}{2} \cdot \ln|1+x^2| + C
\end{aligned}$$

$$\boxed{\boxed{\int \arctan(x) dx = x \arctan(x) - \frac{1}{2} \cdot \ln(1+x^2) + C}}$$

f)  $\int e^x \cos(x) dx$

$$\begin{aligned}
u' &= e^x \quad | \quad u = e^x \\
v &= \cos(x) \quad | \quad v' = -\sin(x) \\
&= e^x \cos(x) - \int e^x (-\sin(x)) dx
\end{aligned}$$

$$u' = e^x \quad | \quad u = e^x$$

$$v = -\sin(x) \quad | \quad v' = -\cos(x)$$

$$\begin{aligned} &= e^x \cos(x) - (e^x(-\sin(x))) - \int e^x(-\cos(x)) dx \\ &= e^x \cos(x) - e^x \sin(x) - \int e^x \cos(x) dx \quad | + \int e^x \cos(x) dx \end{aligned}$$

$$2 \int e^x \cos(x) dx = e^x \cos(x) - e^x \sin(x)$$

$$2 \int e^x \cos(x) dx = e^x(\cos(x) - \sin(x)) \quad | \cdot \frac{1}{2}$$

$$\boxed{\boxed{\int e^x \cos(x) dx = \frac{1}{2}e^x(\cos(x) - \sin(x)) + C}}$$

g)  $\int \cos(\ln(x)) dx$

$$\int \cos(\ln(x)) dx = \int 1 \cdot \cos(\ln(x)) dx$$

$$u' = 1 \quad | \quad u = x$$

$$v = \cos(\ln(x)) \quad | \quad v' = -\sin(\ln(x)) \cdot \frac{1}{x} \quad | \text{Kjernereg}$$

$$= x \cos(\ln(x)) - \int x(-\sin(\ln(x)) \cdot \frac{1}{x}) dx$$

$$= x \cos(\ln(x)) + \int \sin(\ln(x)) dx$$

$$u' = 1 \quad | \quad u = x$$

$$v = \sin(\ln(x)) \quad | \quad v' = \cos(\ln(x)) \cdot \frac{1}{x} \quad | \text{Kjernereg}$$

$$= x \cos(\ln(x)) + (x \sin(\ln(x)) - \int x(\cos(\ln(x)) \cdot \frac{1}{x}) dx$$

$$= x \cos(\ln(x)) + x \sin(\ln(x)) - \int \cos(\ln(x)) dx \quad | + \int \cos(\ln(x)) dx$$

$$2 \int \cos(\ln(x)) dx = x \cos(\ln(x)) + x \sin(\ln(x))$$

$$2 \int \cos(\ln(x)) dx = x(\cos(\ln(x)) + \sin(\ln(x))) \quad | \cdot \frac{1}{2}$$

$$\boxed{\boxed{\int \cos(\ln(x)) dx = \frac{1}{2}x(\cos(\ln(x)) + \sin(\ln(x))) + C}}$$

## 1.5 Oppgave 5

Regn ut følgende ubestemte integraler ved hjelp av delbrøkkopp spalting:

a)  $\int \frac{3x+3}{x^2-9} dx$

Finner rot til nevner.

$$x^2 - 9 = 0 \quad | +9$$

$$x^2 = 9 \quad | \sqrt{\phantom{x}}$$

$$x = \pm\sqrt{9} = \pm 3$$

Faktoriserer nevner.

$$x^2 - 9 = xx - 3 \cdot 3 = (x + 3)(x - 3)$$

Delbrøkkopp spalting.

$$\frac{3x+3}{x^2-9} = \frac{A}{x+3} + \frac{B}{x-3} \quad | \cdot (x+3)(x-3)$$

$$3x+3 = A(x-3) + B(x+3)$$

Finner  $A$  og  $B$  ved å definere  $x$ .

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$$x = -3$$

$$3 \cdot (-3) + 3 = A((-3) - 3) + B((-3) + 3)$$

$$-6 = -6A + 0 \quad | : -6$$

$$A = 1$$


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$$x = 3$$

$$3 \cdot 3 + 3 = A(3 - 3) + B(3 + 3)$$

$$12 = 0 + 6B \quad | : 6$$

$$B = 2$$


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$$\frac{3x+3}{x^2-9} = \frac{1}{x+3} + \frac{2}{x-3}$$

Kontrollerer ved å addere brøker.

$$\frac{x-3}{x-3} \cdot \frac{1}{x+3} + \frac{x+3}{x+3} \cdot \frac{2}{x-3} = \frac{x-3+2x+6}{(x+3)(x-3)} = \frac{3x+3}{x^2-9}$$

Integrer uttrykket.

$$\begin{aligned} \int \frac{3x+3}{x^2-9} dx &= \int \left( \frac{1}{x+3} + \frac{2}{x-3} \right) dx \\ &= \int \frac{1}{x+3} dx + 2 \int \frac{1}{x-3} dx = \ln|x+3| + 2\ln|x-3| + C \\ &= \ln|x+3| + 2\ln|x-3| + C \\ &= \ln((x+3)(x-3)^2) + C \end{aligned}$$

$$\boxed{\int \frac{3x+3}{x^2-9} dx = \ln((x+3)(x-3)^2) + C}$$

b)  $\int \frac{x^2+x+1}{x^3+x} dx$

Ekspanderer brøken.

$$\frac{x^2+x+1}{x^3+x} = \frac{x^2}{x^3+x} + \frac{x}{x^3+x} + \frac{1}{x^3+x}$$

$$\int f(x) + g(x) dx = \int f(x) dx + \int g(x) dx$$

$$\int \frac{x^2+x+1}{x^3+x} dx = \int \frac{x^2}{x^3+x} dx + \int \frac{x}{x^3+x} dx + \int \frac{1}{x^3+x} dx$$

Integrer uttrykket.

$$\int \frac{x^2}{x^3+x} dx \quad | \quad x^3+x = x(x^2+1)$$

$$\int \frac{x^2}{x(x^2+1)} dx$$

$$\int \frac{x}{x^2+1} dx$$

$$u = x^2 + 1$$

$$\frac{du}{dx} = 2x \quad | \quad \cdot \frac{dx}{2x}$$

$$\frac{1}{2x} du = dx$$

$$\int \frac{x}{u} \cdot \frac{1}{2x} du$$

$$\int \frac{1}{u} \cdot \frac{1}{2} du$$

$$\frac{1}{2} \int \frac{1}{u} du \quad | \quad \int \frac{1}{x} dx = \ln|x| + C$$

$$\frac{1}{2} \ln(x^2+1) + C$$

$$\int \frac{x}{x^3+x} dx = \int \frac{x}{x(x^2+1)} dx = \int \frac{1}{x^2+1} dx = \arctan(x) + C$$

$$\int \frac{1}{x^3+x} dx = \int \frac{1}{x(x^2+1)} dx$$

$$\frac{1}{x(x^2+1)} = \frac{A}{x} + \frac{Bx+C}{x^2+1} = \frac{A}{x} + \frac{Bx}{x^2+1} + \frac{C}{x^2+1} \quad | \cdot x(x^2+1)$$

$$A(x^2+1) + B(x^2) + C(x) \quad \star$$

Prøver med  $x = 0$ .

$$1 = A(0^2+1) + B(0^2) + C(0)$$

$$A = 1$$

Prøver med  $x = 1$ .

$$1 = 1(1^2 + 1) + B(1^2) + C(1)$$

$$1 = 2 + B + C$$

$$B = C - 1$$

Prøver med  $x = 2$  med hensyn på  $C$ .

$$1 = 1(2^2 + 1) + (C - 1)2^2 + C2$$

$$1 = 5 + 4C - 4 + 2C$$

$$0 = 6C$$

$$C = 0$$

Prøver med  $x = 3$  med hensyn på  $B$ .

$$1 = 1(3^2 + 1)B(3^2) + 0 \cdot 3$$

$$1 = 10 + B9$$

$$-B9 = 9 \quad |: -9$$

$$Bx = -1$$

$$A = 1, \quad Bx = -1, \quad C = 0$$

$$\frac{1}{x(x^2 - 1)} = \frac{1}{x} - \frac{x}{x^2 - 1} + \frac{0}{x^2 - 1}$$

$$\int \frac{1}{x(x^2 - 1)} dx = \int \frac{1}{x} dx - \int \frac{x}{x^2 - 1} dx$$

$$\int \frac{1}{x} dx = \ln|x| + C$$

$$- \int \frac{x}{x^2 - 1} dx = \int \frac{x}{x^2 + 1} dx$$

$$u = x^2 + 1$$

$$\frac{du}{dx} = 2x \quad | \cdot \frac{dx}{2x}$$

$$\frac{1}{2x} du = dx$$

$$\int \frac{x}{u} \cdot \frac{1}{2x} du = \frac{1}{2} \int \frac{1}{u} du = \frac{1}{2} \ln|u| + C = \frac{1}{2} \ln(x^2 + 1) + C$$

$$\int \frac{x^2 + x + 1}{x^3 + x} dx = \frac{1}{2} \ln(x^2 + 1) + \arctan(x) + \ln|x| - \frac{1}{2} \ln(x^2 + 1) + C$$

$$\boxed{\int \frac{x^2 + x + 1}{x^3 + x} dx = \arctan(x) + \ln|x| + C}$$